

Mixture of Experts in LLM

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- 1 Structure of MoE
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Basic Setting of MoE Layer

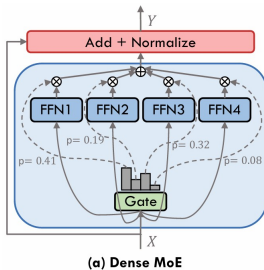
In transformer-based LLMs, each mixture-of-experts (MoE) layer consists of

- N expert networks $\{f_1, \dots, f_N\}$. Each f_i is parameterized by W_i , accepting the same input $\mathbf{x}$ and generating an output $f_i(\mathbf{x}; W_i)$.
- A gating network $\mathcal{G}$ which usually takes the form of a linear network with a softmax activation function and is parameterized by Θ , directs the input to appropriate expert networks through its output $\mathcal{G}(\mathbf{x}; \Theta)$.

Based on the design of gating function $\mathcal{G}$, MoE layers can be broadly classified into two categories: dense MoE and sparse MoE.

Dense MoE

The dense MoE layer activates all f_i s during each iteration.



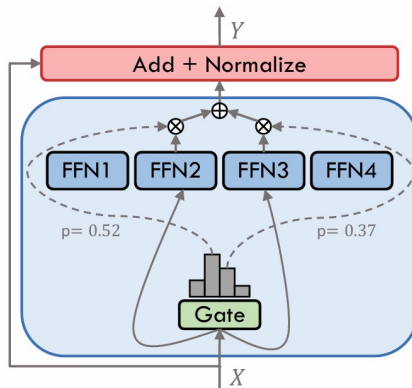
$$\mathcal{F}_{\text{dense}}^{\text{MoE}}(\mathbf{x}; \Theta, \{W_i\}_{i=1}^N) = \sum_{i=1}^N \mathcal{G}(\mathbf{x}; \Theta)_i f_i(\mathbf{x}; W_i),$$

$$\mathcal{G}(\mathbf{x}; \Theta)_i = \text{softmax}(g(\mathbf{x}; \Theta))_i = \frac{\exp(g(\mathbf{x}; \Theta)_i)}{\sum_{j=1}^N \exp(g(\mathbf{x}; \Theta)_j)},$$

where $g(\mathbf{x}; \Theta)$ represents the gating value prior to the softmax operation.

Sparse MoE

- High computational overhead in dense MoE.
- Activate only a selected subset of experts during each forward pass. Compute a weighted sum of the outputs of only the top- k experts.



(b) Sparse MoE

Sparse MoE

$$\mathcal{G}(\mathbf{x}; \Theta)_i = \text{softmax}(\text{TopK}(g(\mathbf{x}; \Theta) + \mathcal{R}_{\text{noise}}, k))_i,$$

$$\text{TopK}(g(\mathbf{x}; \Theta), k) := \begin{cases} g(\mathbf{x}; \Theta)_i, & \text{condition,} \\ -\infty, & \text{otherwise.} \end{cases}$$

condition: $g(\mathbf{x}; \Theta)_i$ is in the top- k elements of $g(\mathbf{x}; \Theta)$.

- The hyper-parameter k is selected based on the specific application, with common choices being $k = 1$ or 2 .
- The noise $\mathcal{R}_{\text{noise}}$ encourages exploration among experts, enhancing the stability of MoE training.

Sparse MoE: Load-Balancing

The uneven distribution of workload across experts will lead to a load balancing issue. Consider a batch of queries $\mathcal{B} = \{\mathbf{x}_1, \dots, \mathbf{x}_T\}$ comprising T tokens. Let $k = 1$. Define the auxiliary load balancing loss for the batch as

$$\begin{aligned}\mathcal{L}_{\text{load-balancing}} &= N \sum_{i=1}^N \mathcal{D}_i \mathcal{P}_i; \\ \mathcal{D}_i &= \frac{1}{T} \sum_{\mathbf{x} \in \mathcal{B}} \mathbb{I}\{\arg \max \mathcal{G}(\mathbf{x}; \Theta) = i\}, \\ \mathcal{P}_i &= \frac{1}{T} \sum_{\mathbf{x} \in \mathcal{B}} \mathcal{G}(\mathbf{x}; \Theta)_i.\end{aligned}$$

- $\mathcal{D}_i$ represents the proportion of tokens distributed to expert f_i
- $\mathcal{P}_i$ denotes the mean gating probability assigned to f_i

To ensure an even distribution of the batch of tokens across the N experts, $\mathcal{L}_{\text{load-balancing}}$ should be minimized.

Effect of MoE

- Leverage a collection of specialized models and a gating mechanism to select the appropriate “expert networks”
- Effectively scale model parameters without a corresponding increase in computational demand

The incorporation of MoE architectures into LLMs enables these models to achieve significant parameter scale-ups and consequent enhancements in capabilities.

- DeepSeekMoE 16B, while requiring approximately 40% less computation, attains performance on par with the Llama 2 7B.
- Qwen1.5-MoE with only 2.7B active parameters rivals the performance of leading 7B parameter models such as the Mistral 7B and the Qwen1.5-7B.

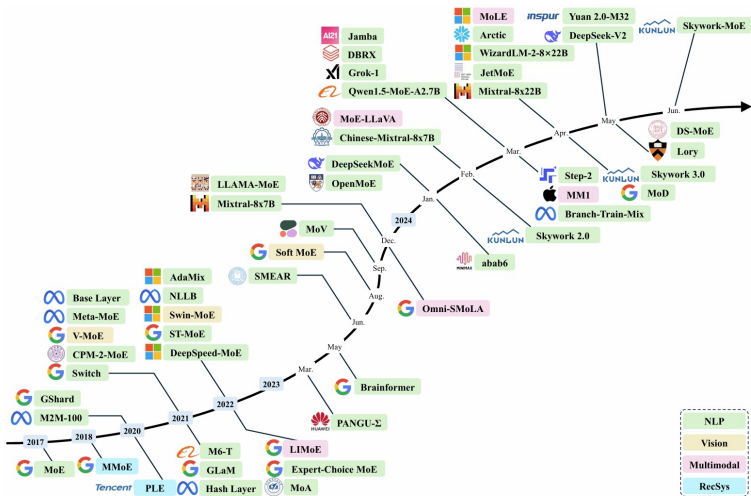


Fig. 1. A chronological overview of several representative mixture-of-experts (MoE) models in recent years. The timeline is primarily structured according to the release dates of the models. MoE models located above the arrow are open-source, while those below the arrow are proprietary and closed-source. MoE models from various domains are marked with distinct colors: Natural Language Processing (NLP) in green, Computer Vision in yellow, Multimodal in pink, and Recommender Systems (RecSys) in cyan.

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Gating (Router) Function

Three distinct gating mechanism:

- Sparse: activate a subset of experts,
- Dense: activate all experts,
- Soft: encompass fully differentiable approaches including input token merging and expert merging.

Sparse Gating: Training

- Sparse gating may be trained by various forms of reinforcement learning and back-propagation, making binary or sparse and continuous, stochastic or deterministic gating decision
- Shazeer: A differentiable heuristic with auxiliary load balancing losses [8], introducing a differentiable aspect to the gating process, thereby facilitating the derivation of gradients that can guide the gating function's optimization.
- Due to its selection of experts for each input token, this method can be recognized as a gating function with token choice.

Token-Choice Gating: Top-K

- Shazeer [8]: Top-k experts, with $k > 1$, to enhance the efficacy of MoE.

Increasing the value of k enhances performance, which has led to the widespread adoption of this top-k strategy with $k > 1$.

- Switch Transformer model [2] has shown that top-1 gating strategy can also yield competitive results.

M6-t [9]: expert prototyping, a variation of the top-1 gating.

Organizes experts into k groups and then applies top-1 gating in each group.

Auxiliary Loss for Token-Choice Gating

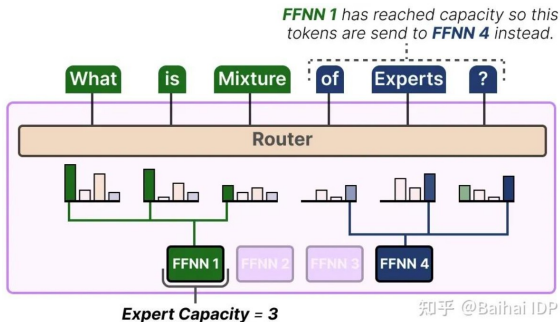
If too many input tokens are assigned to a certain expert network, some expert networks may not be fully trained.

Token-choice gating incorporates an auxiliary loss to promote equitable token distributions across experts.

- Shazeer: $\mathcal{L}_{\text{importance}} + \mathcal{L}_{\text{load}}$ [8]. $\mathcal{L}_{\text{importance}}$, the square of the coefficient of variation of the set of importance values and multiplied by a hand-tuned scaling factor, promotes balance in importance among experts. $\mathcal{L}_{\text{load}}$ guarantees an even distribution of training examples among experts.
- GShard: $\mathcal{L}_{\text{aux}}$ [3], a differentiable approximation for the loss (the dot-product of mean gates and mean gating decisions per expert), becomes paradigm for later models.
- Mod-Squad: $\mathcal{L}_{\text{MI}}$ [1] maximize the mutual information(MI) between experts and tasks to build task-expert alignment, enhancing model's ability for multi-task.

Expert Capacity

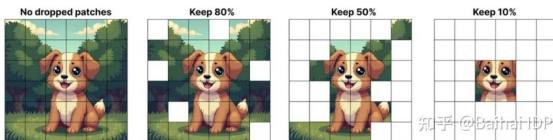
- GShard [3] also introduces expert capacity limits to control token load.
 - Select a secondary expert



- If all relevant expert networks have reached their processing capacity, then these tokens will not be processed by any expert network, leading to token overflow.

Routing improvement

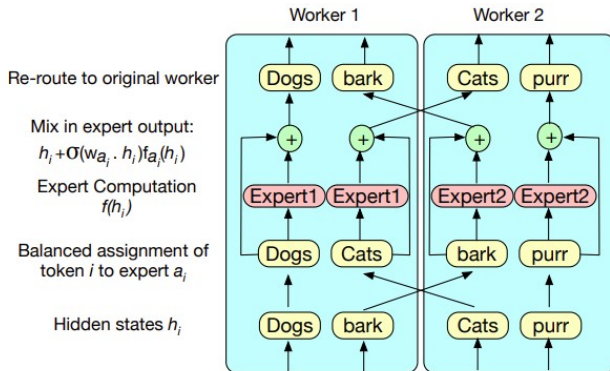
- Batch Prioritized Routing (BPR)
 - Proposed for image classification tasks with Vision Transformer (ViT) models [7].
 - Prioritizes tokens with higher gating scores over sequence order.



- Token Bias Mitigation: Randomize within-sequence token priorities to reduce early-position routing bias.

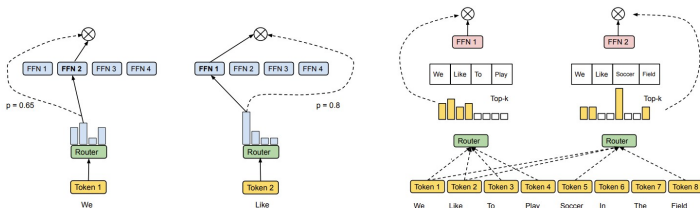
Balanced Assignment of Sparse Experts

- [4] casts the token-to-expert allocation process as a linear assignment problem
- Maximize the token-expert affinities under the constraints that each expert is assigned an equal quantity of tokens.



Expert-Choice Sparse Gating

- Each expert selects the top-k tokens they will process [12].



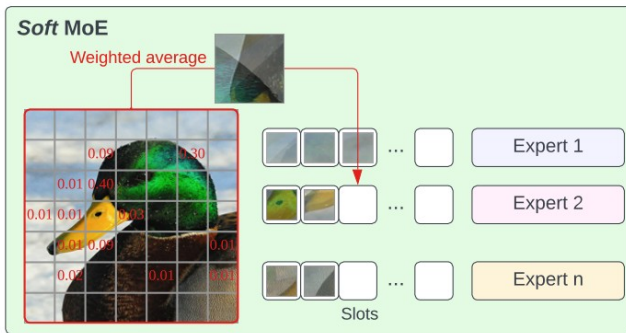
- Ensure a uniform distribution of tokens across experts.
- Offer an adaptive computational interpretation where the model can implicitly apply more computation to certain tokens.

From Sparse to Soft

- Sparse MoE often necessitates heuristic auxiliary losses to ensure balanced expert engagement and to minimize unassigned tokens.
- Soft MoE approach maintains full differentiability by leveraging all the experts for each input, avoiding issues inherent to discrete expert selection.
- Soft MoE apply the gating-weighted merging of tokens or experts to reduce computational cost.

Token Merging

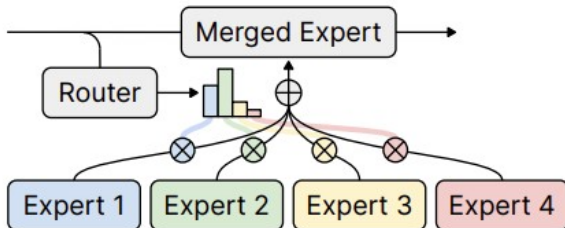
- Puigcerver [6] computes weighted averages of all tokens,
- The weights depends on both tokens and experts
- Processes each aggregate with its respective expert.



Expert Merging: SMEAR

Soft Merging of Experts with Adaptive Routing (SMEAR)

- [5] merge all the experts' parameters through a weighted average.

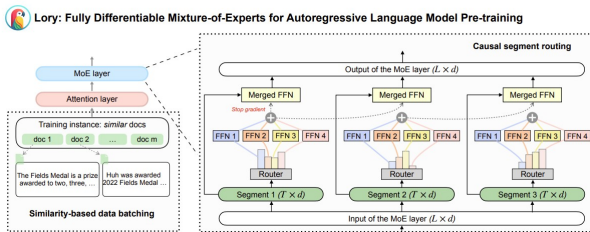


- Not incur a significant increase in computational costs and enables standard gradient-based training.

Expert Merging: Lory

Lory [11]: for auto-regressive language model pre-training

- Causal segment routing strategy: conduct expert merging at the segment level and preserves the auto-regressive property of language models.
- Similarity-based data batching: construct training instances, steer the experts toward specializing in specific domains

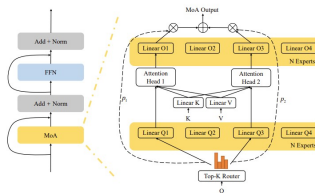


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Network Types

- MoE serves as a substitute for Feed-Forward Network (FFN) modules, because FFN layers exhibit higher sparsity and more domain specificity than self-attention layers.
- Mixture of Attention Heads (MoA) [10]



- Two sets of experts: one for query projection, one for output projection.
- Share the key and value projection weights across attention experts.

Hyperparameters in MoE

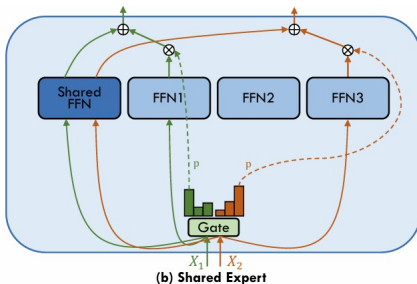
- Expert Count
 - GLaM: a 64 expert setup with top-2 gating strikes an optimal balance between execution efficiency and performance.
 - Recent sparse MoE models commonly utilize no more than 64 experts.
- Expert Size
 - GLaM prioritizes the expansion of the intermediate hidden dimension per expert to scale the model effectively
 - DeepSeekMoE: fine-grained expert segmentation—subdivide the intermediate hidden dimension of FFN expert, while preserve the overall parameter count.
 - LLAMA-MoE: activating 4 out of 16 experts with $d_{expert}=688$ marginally outperforms the activation of 2 out of 8 experts with $d_{expert}=1376$

Hyperparameters in MoE

- Frequency of MoE Layers
 - Typically intersperse MoE layers in place of the dense FFN layers. Most MoE models features alternate FFN replacement ($1/2$) with MoE layers.
 - After Mixtral $8 \times 7B$, the trend seems to place MoE in every layer of the model, with a common choice of only 8 or 16 experts.
 - Different insertion strategies have different effects on performance and speed.

Shared Experts

- Residual MoE: each token is processed by a fixed expert and another selected by gating, achieving two experts engagement per layer without increasing the communication cost.
- The gating-selected MoE expert serves as an error-correcting to the fixed dense FFN.
- This paradigm of integrating fixed FFN with sparse MoE, often referred to as shared expert.



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MoPE

- PEFT (parameter-efficient fine-tuning): only update a small set of parameters while maintaining the rest of the base model untouched.
- However, PEFT approaches often struggle with generalizing across multiple tasks due to their limited scope of trainable parameters and the potential for catastrophic forgetting.
- MoPE (mixture of parameter-efficient experts): each expert constructed using PEFT techniques. MoPE operates with fewer parameters, achieving greater resource efficiency compared to traditional MoE models.

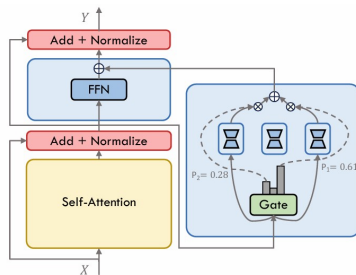
Feed-Forward Network

During the training, focus on optimizing the parameter-efficient experts and the gating mechanism, leaving the rest of the pre-trained model intact.

$$\text{FFN}^{\text{MoE}}(x') = \text{FFN}(x') + \sum_{i=1}^n x' \Delta W_i^{\text{ffn}} G^{\text{ffn}}(x')_i$$

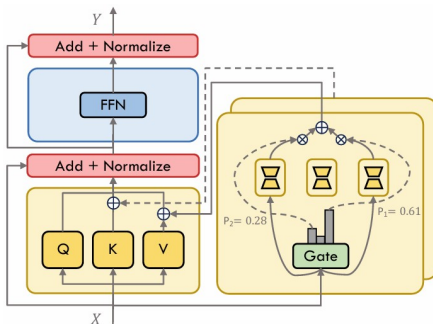
$$x' = \text{LayerNorm}(\text{SA}(x) + x),$$

where $\Delta W^{\text{ffn}}, G^{\text{ffn}}(x)$ is the parameter-efficient expert and gating function applied to the FFN layer.



Attention

- MoPE framework can be applied to the Query, Key, Value, and Output projection modules, individually or in various combinations.
- During fine-tuning, only the parameters of the activated experts and the gating network are updated, while the remaining parameters of the model are kept frozen.



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